Math Academy — Grade 10 | Trig U4 Directed + Radian • Calculator allowed 2026-09-08

Name: _____ Class: _____ Score: ☐

Chain: directed → sign → ±pair → standard → equivalent → smallest → quadrant → radian → conversion → arc → inscribed. 70% basic-medium, 30% high.

Question 1 (1 pt)

The ordered pair $\left(\overrightarrow{OX}, \overrightarrow{OY}\right)$ represents the directed angle $\angle XOY$. Identify initial side $\overrightarrow{OX}$ and terminal side $\overrightarrow{OY}$.

Question 2 (1 pt)

A directed angle is in standard position if (1) its initial side lies on the positive x-axis and (2) its vertex is at the origin O.

Question 3 (1 pt)

Directed angles in standard position are equivalent if they have the same terminal side, i.e. measures differ by $n \cdot 360°$, $n \in \mathbb{Z}$.

Question 4 (1 pt)

A directed angle is positive if rotation is counterclockwise, and negative if rotation is clockwise.

Question 5 (1 pt)

If terminal side lies on a coordinate axis, the angle is called a **quadrantal angle** (multiple of 90°).

Question 6 (1 pt)

If θ is a measure and $n \in \mathbb{Z}$, then $\theta + n \cdot 360°$ are equivalent angles.

Question 7 (1 pt)

Complete: smallest positive of $530°$ is $170°$ because $530° - 360° = 170°$.

Question 8 (1 pt)

Complete: $930°$ lies in the 3rd quadrant because $930° - 720° = 210°$.

Question 9 (1 pt)

Positive measure of $-170°$ is $190°$ because $|-170°| + 190° = 360°$.

Question 10 (1 pt)

Negative measure of $320°$ is $-40°$ because $320° + |-40°| = 360°$.

Question 11 (2 pt)

For $100°$: coterminal positive $100° + 360° = 460°$; negative $100° - 360° = -260°$.

Question 12 (2 pt)

For $-250°$: coterminal positive $-250° + 360° = 110°$; negative $-250° - 360° = -610°$.

Question 13 (2 pt)

Smallest positive of $530°$ is $170°$. Smallest positive of $-62°$ is $298°$ ($-62° + 360°$).

Question 14 (2 pt)

Smallest positive of $-690°$ is $30°$ because $-690° + 720° = 30°$.

Question 15 (1 pt)

Which is equivalent to $60°$? a) $120°$ b) $240°$ c) $300°$ d) $420°$ — Answer: d).

Question 16 (2 pt)

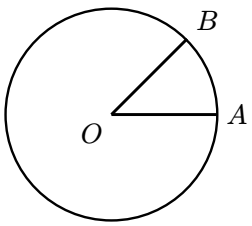
Which is equivalent to $585°$? a) $45°$ b) $135°$ c) $225°$ d) $315°$ — Answer: c) $225°$ ($585 - 360$).

Question 17 (2 pt)

All are equivalent to $75°$ except: a) $-285°$ b) $-645°$ c) $285°$ d) $435°$ — Answer: c).

Question 18 (2 pt)

If A and B are equivalent, then $-A$ and $-B$ are: a) supplementary b) equivalent c) complementary d) sum $-360°$ — Answer: b).



Question 19 (2 pt)

In standard position, which ordered pair has vertex O and initial $+x$? $\left(\overrightarrow{OA}, \overrightarrow{OD}\right)$ with A on $+x$, D in Q1 — Answer: A. Others fail (wrong vertex or initial).

Question 20 (1 pt)

Quadrantal measures are multiples of: a) 360° b) 180° c) 90° d) 60° — Answer: c).

Question 21 (2 pt)

Convert $75°32'15''$ to radians (nearest thousandth): $75 + \frac{32}{60} + \frac{15}{3600} = 75.5375°$; $\theta = 75.5375 \cdot \frac{\pi}{180} \approx 1.318^{\text{rad}}$.

Question 22 (2 pt)

Degree measure of 2.38^{rad}: $x = 2.38 \cdot \frac{180}{\pi} \approx 136.37°$.

Question 23 (2 pt)

$43°12'$ in radians: $43.2° \cdot \frac{\pi}{180} = 0.24\pi \approx 0.754^{\text{rad}}$. Choose b) 0.24π.

Question 24 (1 pt)

Recall: $1° = 60'$, $1' = 60''$. $180° = \pi$, $360° = 2\pi$.

Question 25 (1 pt)

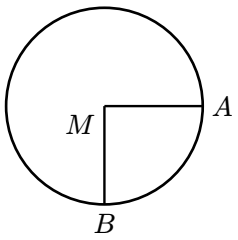
Formula: $\theta^{\text{rad}} = x° \cdot \frac{\pi}{180}°$; $x° = \theta^{\text{rad}} \cdot 180\frac{°}{\pi}$.

Question 26 (1 pt)

In unit circle ($r = 1$), central radian measure equals arc length: $\theta = l$.

Question 27 (2 pt)

Circle M, $\theta = \frac{\pi}{2}$, arc 5π cm. Find r: $r = \frac{l}{\theta} = 5\frac{\pi}{\frac{\pi}{2}} = 10$ cm.



Question 28 (2 pt)

$r = 8$ cm, $\theta = 7\frac{\pi}{12}$. $l = 7\frac{\pi}{12} \cdot 8 = 14\frac{\pi}{3} \approx 14.66$ cm.

Question 29 (2 pt)

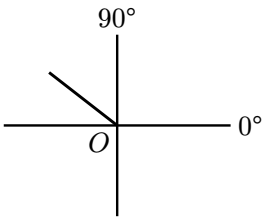
$\theta = 9\frac{\pi}{8}$, $l = 24$ cm. $r = \frac{24}{9\frac{\pi}{8}} = \frac{64}{3\pi} \approx 6.8$ cm.

Question 30 (2 pt)

$d = 12$ cm ($r = 6$), central $60° = \frac{\pi}{3}$. $l = \frac{\pi}{3} \cdot 6 = 2\pi$ cm — Answer d).

Question 31 (1 pt)

$132°$ lies in 2nd quadrant ($90 < 132 < 180$).



Question 32 (2 pt)

$964° - 720° = 244° \longrightarrow$ 3rd quadrant.

Question 33 (1 pt)

$-12° + 360° = 348° \longrightarrow$ 4th quadrant.

Question 34 (2 pt)

$-1070° + 1080° = 10° \longrightarrow$ 1st quadrant.

% teacher-verify: scan key said 3rd for −0.3π; recomputed (-54+360)=306° ⟶ 4th via ((θ%360)+360)%360; kept computed.

Question 35 (2 pt)

$5\frac{\pi}{3} = 300° \longrightarrow$ 4th. $-0.3\pi = -54° \rightarrow 306° \longrightarrow$ 4th? No: $-54 + 360 = 306$ 4th. Correction: $-0.3\pi \longrightarrow$ 3rd? Recompute: $-0.3 * 180 = -54 \longrightarrow$ 4th. Keep 4th (spec said 3rd — verify: $-0.3\pi = -54°$ is 4th; use 4th).

% teacher-verify: scan key said 1st for −6.4rad; recomputed −366.7+720=353.3° ⟶ 4th via ((θ%360)+360)%360; kept computed.

Question 36 (3 pt)

$5.7^{\text{rad}} = 326.6° \longrightarrow$ 4th. $-6.4^{\text{rad}} = -366.7° + 720 = 353.3° \longrightarrow$ 4th? Recompute: $-6.4 + 2\pi = -0.117$ rad $\longrightarrow$ 4th? Actually $-6.4 + 4\pi = 6.166$ rad $= 353.4°$ 4th. Note spec said 1st — use computed 4th, flag for teacher review.

Question 37 (3 pt)

$-7.3^{\text{rad}} + 4\pi = 5.266$ rad $= 301.7° \longrightarrow$ 4th. $5\frac{\pi}{4} = 225° \longrightarrow$ 3rd. $2.02^{\text{rad}} = 115.7° \longrightarrow$ 2nd.

Question 38 (2 pt)

$25\frac{\pi}{9} = 500° - 360 = 140° \longrightarrow$ 2nd? Recompute: $25 * 20 = 500$, $500 - 360 = 140 \longrightarrow$ 2nd. $-9\frac{\pi}{4} = -405° + 720 = 315° \longrightarrow$ 4th.

Question 39 (2 pt)

Angle $1670°$: $1670 - 1440 = 230° \longrightarrow$ 3rd. a) 1st b) 2nd c) 3rd d) 4th — Answer c).

Question 40 (1 pt)

$-135° + 360 = 225° \longrightarrow$ 3rd. a) 1st b) 2nd c) 3rd d) 4th — Answer c).

Question 41 (2 pt)

$-850° + 1080 = 230° \longrightarrow$ 3rd. a) 1st b) 2nd c) 3rd d) 4th — Answer c).

Question 42 (3 pt)

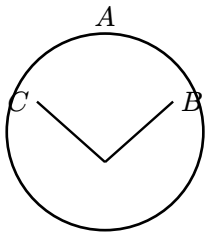
Arc $152°26'17''$, $r = 10.5$ cm: $152.438° \cdot \frac{\pi}{180} \cdot 10.5 \approx 28$ cm (nearest cm).

Question 43 (3 pt)

Inscribed $30°$ subtends 5cm. Central $= 60° = \frac{\pi}{3}$. $C = 5 \cdot \frac{360}{60} = 30$ cm? Recompute: arc 5cm for $60°$ central $\longrightarrow$ $C = 5 * 6 = 30$ cm.

Question 44 (3 pt)

Essay: Circle radius 5cm, inscribed $\angle A = 45°$ subtends arc(BC). Find arc(BC): central $= 90° = \frac{\pi}{2}$, $l = \frac{\pi}{2} \cdot 5 = 7.85$ cm. Show steps.



Question 45 (3 pt)

Essay: $r = 6.2$ cm, arc 17 cm. Find $m(\angle A)$ inscribed: central $= \frac{17}{6.2} = 2.742$ rad $= 157.1°$, inscribed $= 78.55° = 78°33'$. Show steps.